Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 07)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 1 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 1 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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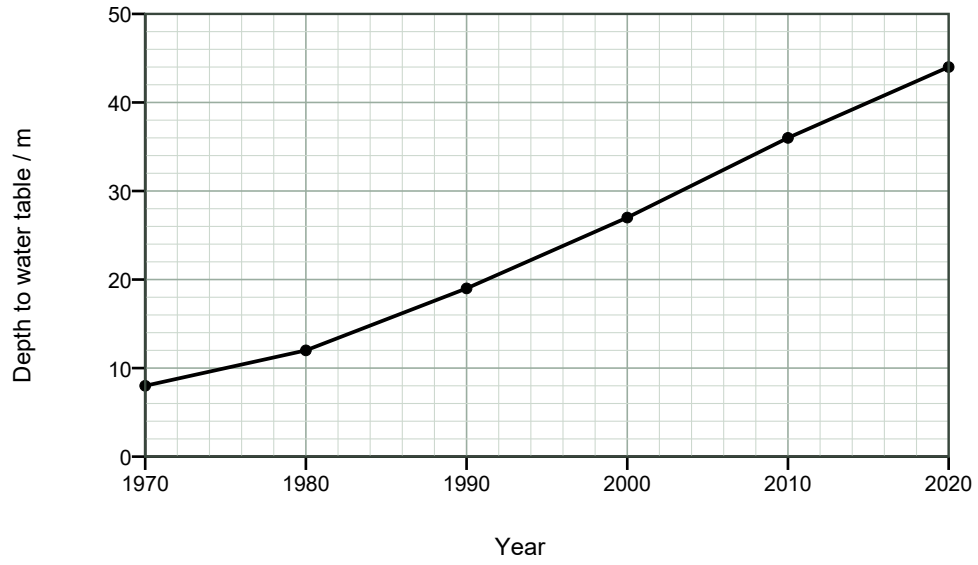
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0 2 Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1



0 2 . 1 Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]

Answer ____ m/yr

0 2 . 2 Explain why the water table is falling.

[2 marks]

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0 2 . 3 Explain the features a rock must have to form an aquifer.

[3 marks]

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0 2 . 4 Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

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0 3 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 3 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 3 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 4 . 1 Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

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0 4 . 2 Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

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0 4 . 3 Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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0 4 . 4 Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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0 5 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 5 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 5 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 6 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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0 6 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

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Useful properties .....

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Properties causing ozone depletion .....

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0 6 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 6 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 6 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 6 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 7 . 1 Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

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0 7 . 2 Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

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0 7 . 3 Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

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0 7 . 4 Give **two** industrially important gases extracted from the atmosphere.

[2 marks]

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0 8 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

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0 8 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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0 8 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

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0 8 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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0 9 Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	**Cadmium in rice / mg kg^{-1}**	**Cadmium in kidney of residents / mg kg^{-1}**	**Cases of bone disease (itai-itai) per 1000 people**
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

0 9 . 1 Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer ..... times

0 9 . 2 Explain why cadmium and zinc pollution often occur together.

[1 mark]

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0 9 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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0 9 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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0 9 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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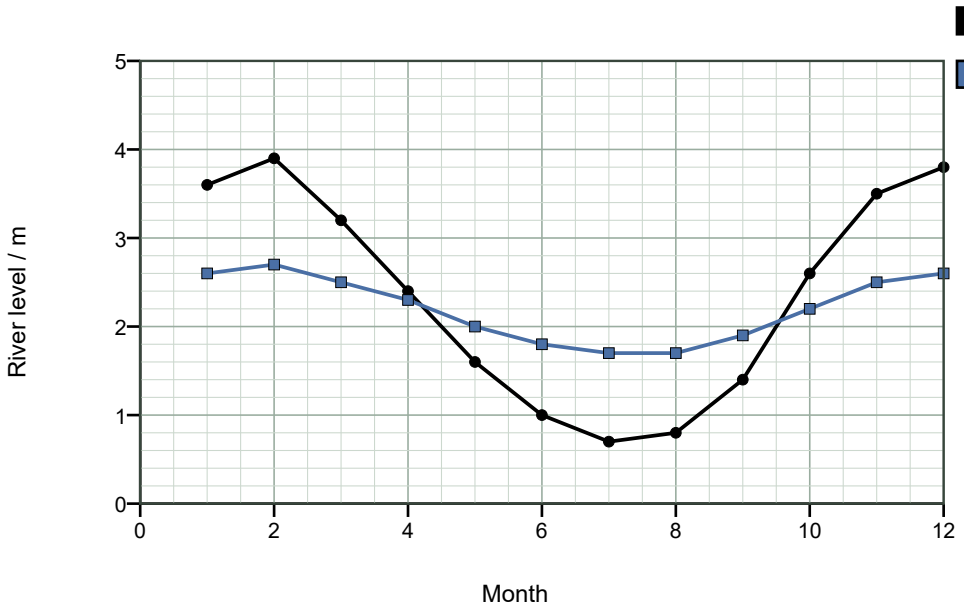
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

11 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

11 . 2 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS